

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Randomized, Phase 3 Trial With Anti-PD-1 Monoclonal Antibody Pembrolizumab (MK-3475) Versus Placebo for Patients With Early Stage NSCLC After Resection and Completion of Standard Adjuvant Therapy (PEARLS)
Short Title:	Study of Pembrolizumab (MK-3475) vs Placebo for Participants With Non-small Cell Lung Cancer After Resection With or Without Standard Adjuvant Therapy (MK-3475-091/KEYNOTE-091)
Protocol Number:	794471016954218372
NCT Number:	NCT02504372
Sponsor Name:	Merck
Investigational Product:	pembrolizumab
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Disease-Free Survival (DFS): DFS was defined as the time from randomization to either the date of disease recurrence or death (whatever the cause) as assessed by the investigator. Recurrence of disease was defined as local regional recurrence, a distant (metastatic) recurrence, or a second primary cancer. Occurrence of a second extra-pulmonary malignancy was considered to be an event. 2. DFS in Programmed Death Ligand-1 (PDL-1) Strong Positive Participants With Tumor Proportion Score (TPS) ≥50%: DFS in PDL-1 strong positive participants with TPS ≥50% was defined as the time from randomization to either the date of disease recurrence or death (whatever the cause) as assessed by the investigator. Recurrence of disease was defined as local regional recurrence, a distant (metastatic) recurrence, or a second primary cancer. Occurrence of a second extra-pulmonary malignancy was considered to be an event.
Secondary Obj.:	1. DFS in PDL-1 Strong Positive Participants With TPS ≥1% 2. Overall Survival (OS) 3. OS in PDL-1 Strong Positive Participants With TPS ≥50%

2.2 Background & Rationale

In this study, participants with Stage IB/II-IIIA non-small cell lung cancer (NSCLC) who have undergone surgical resection (lobectomy or pneumonectomy) with or without adjuvant chemotherapy will be treated with pembrolizumab or placebo. The primary study hypothesis is that pembrolizumab will provide improved disease-free survival (DFS) versus placebo.

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE3, Status: ACTIVE_NOT_RECRUITING
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Target N: 1177

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Pathological diagnosis of NSCLC confirmed at surgery, any histology. Participants with two synchronous primary non-small cell lung cancers are excluded from the study
- * Union for International Cancer Control (UICC) v7 Stage IB with T ≤ 4 cm, II-IIIa NSCLC after complete surgical resection with resection margins proved microscopically free of disease (R0). Carcinoma in situ can be present at the bronchial margin
- * Available tumor sample obtained at surgical resection for programmed cell death ligand-1 (PD-L1) Immunohistochemistry (IHC) expression assessment
- * Eastern Cooperative Oncology Group (ECOG) Performance Status 0-1
- * Adequate organ function performed within 10 days of treatment initiation
- * Female participants of childbearing potential must have a negative urine or serum pregnancy test at screening (within 72 hours of first infusion of study medication). If the urine test cannot be confirmed as negative, a serum pregnancy test will be required. The serum pregnancy test must be negative for the participant to be eligible
- * Female participants of childbearing potential must be willing to use 2 methods of birth control or be surgically sterile, or abstain from heterosexual activity starting with the first infusion of study treatment through 120 days after the last infusion of study treatment
- * Female participants who are breast feeding must discontinue nursing prior to the first infusion of study medication and until 120 days after the last infusion study treatment
- * Male participants must agree to use an adequate method of contraception starting with the first infusion of study treatment through 120 days after the last infusion of study treatment
- * Absence of severe comorbidities that in the opinion of the Investigator might hamper the participation to the study and/or the treatment administration
- * No prior or planned neo-adjuvant or adjuvant radiotherapy and/or neo-adjuvant chemotherapy for the current malignancy is allowed

4.2 Exclusion Criteria

- * Evidence of disease at clinical examination and/or baseline radiological assessment as documented by contrast enhanced chest/upper abdomen CT scan, brain CT/MRI and clinical examination
- * More than 4 cycles of adjuvant therapy
- * Prior treatment with anti-programmed cell death (anti-PD)-1, anti-PD ligand-1/2, anti-CD137, or cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) modulators or any other immune-modulating agents
- * Live vaccine within 30 days prior to the first infusion of study treatment
- * Current participation or treatment with an investigational agent or use of an investigational device within 4 weeks of the first infusion of study treatment
- * History of Human Immunodeficiency Virus (HIV) (known HIV 1/2 antibodies positive). No known active Hepatitis B or C
- * Chronic use of immunosuppressive agents and/or systemic corticosteroids or any use in the last 3 days prior to the first infusion of study treatment

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- * History of interstitial lung disease or (non-infectious) pneumonitis that required oral or IV steroids (other than COPD exacerbation) or current pneumonitis
- * Active autoimmune disease that has required systemic treatment in past 2 years
- * History of a hematologic or primary solid tumor malignancy, unless in remission for at least 5 years with the exception of pT1-2 prostatic cancer Gleason score ≤ 6 , superficial bladder cancer, non melanomatous skin cancer or carcinoma in situ of the cervix
- * Previous allogeneic tissue/solid organ transplant
- * Active infection requiring therapy
- * Surgery- or chemotherapy-related toxicity (non-hematological) not resolved to Grade 1 with the exception of alopecia, fatigue, neuropathy and lack of appetite /nausea
- * Pregnant or breastfeeding, or expecting to conceive or father children within the projected duration of the trial, starting with the screening visit through 120 days after the last infusion of study treatment
- * Participant will not be eligible if the participant is or has an immediate family member (e.g., spouse, parent/legal guardian, sibling or child) who is investigational site or Sponsor staff directly involved with this trial, unless prospective site Review Board approval is given allowing exception to this criterion for a specific participant

11. Study Identifiers & Governance

Primary ID: 794471016954218372

Registry Number: NCT02504372

Sponsor: Merck

Study Title: Study of Pembrolizumab (MK-3475) vs Placebo for Participants With Non-small Cell Lung Cancer After Resection With or Without Standard Adjuvant Therapy (MK-3475-091/KEYNOTE-091)

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____